

AI 4403 : AI in Finance

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Financial Literacy

The Four Pillars of Personal Finance

A robust financial foundation requires balancing four critical decisions:

► 1. Saving and Investing (Growth)

- ★ *Saving*: Protecting cash for short-term liquidity
- ★ *Investing*: Taking calculated risks to beat inflation

► 2. Borrowing (Leverage)

- ★ Distinguishing "Good Debt" (Education, Home) from "Bad Debt" (Credit Cards).
- ★ Managing the cost of capital vs. the return on investment.

► 3. Risk Management (Defense)

- ★ Understanding your personal risk appetite and time horizon.
- ★ Diversifying assets to prevent concentrated losses.

► 4. Insurance (Protection)

- ★ Shielding your wealth from catastrophic events (Health & Life cover).
- ★ *Rule of thumb*: Never mix insurance with investments.

Ultimate Goal: Retirement & Legacy

Suppose we have **₹10 lakh**. What would be some simple investment options ?

- ▶ Fixed Deposits
- ▶ Bonds
- ▶ Stock
- ▶ Gold
- ▶ Real Estate
- ▶ Future and Options
- ▶ Basket of Instruments

Question: How to choose between the options ?

Involves answering some critical questions:

- ▶ How should future cash flows be valued ?
- ▶ How should risk be measured ?
- ▶ What constitutes a fair price ?
- ▶ Can risk be reduced or eliminated ?
- ▶ How to fund via borrowing ?

Most people make major financial decisions using **intuition, rules of thumb, or informal advice**.

Financial literacy upgrades your decision-making by providing:

- ▶ **A Quantitative Framework:** To measure and weigh your options objectively.
- ▶ **Tools for Uncertainty:** To transform guesswork into calculated choices.

Math Finance : Helps make decisions objectively

Core Problems in Mathematical Finance

Pricing

What is the fair value of an asset or derivative?

Hedging

How do we perfectly offset or minimize market risks?

Portfolio

How do we optimally allocate capital to maximize return?

The Simplest Investment: Fixed Deposits

An FD is the ultimate "set-and-forget" contract: you lend money to a bank for a guaranteed return.

Example (At 7% Annual Interest):

$$\begin{array}{ccc} \text{₹1,000} & \longrightarrow & \text{₹1,070} \\ \text{Present Value (Today)} & & \text{Future Value (1 Year)} \end{array}$$

Advantages:

- ▶ **Predictable:** Known maturity value.
- ▶ **Low Risk:** Backed by bank guarantees.
- ▶ **Simple:** Minimal decision-making.

Limitations:

- ▶ **No Upside:** Interest rate is locked.
- ▶ **Inflation Risk:** Real returns can be negative.

The Pricing Engine (Discounting)

- ▶ Bank promises ₹1,070 next year at 7% interest rate;
 - ★ What is the *Fair price* of that contract today?
- ▶ Fair Price = $\frac{1,070}{(1+0.07)^1} = 1,000$

The Key Insight:

- ▶ Observe the cash flow
- ▶ Every financial asset (Bonds, Stocks, Options) is priced using cash flows
- ▶ *The only difference? With an FD, the future cash flow is certain*

- ▶ An FD has **one** future cash flow.
- ▶ A Bond has **multiple** future cash flows (coupons + principal).

The Pricing Logic: Simply discount *every* future cash flow and add them together.

The Bond Pricing Engine

$$\text{Fair Price} = \frac{C_1}{(1+r)^1} + \frac{C_2}{(1+r)^2} + \dots + \frac{C_N + \text{Principal}}{(1+r)^N}$$

We are still assuming 100% certainty. No risk models needed yet!

Cashflow in Equity

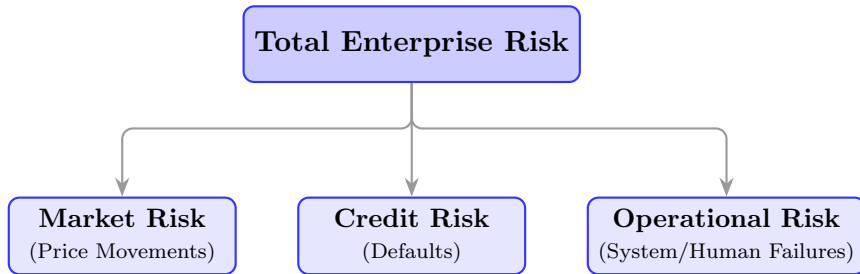
- ▶ With FDs and Bonds, the future cash flows are certain.
- ▶ With a **Equity**, the cash flows (Dividends) are **uncertain**.

The Pricing Formula:

$$\text{Fair Price} = \frac{\mathbb{E}(D_1)}{(1+r)^1} + \frac{\mathbb{E}(D_2)}{(1+r)^2} + \dots$$

Replace certainties with **Expectations**

Uncertainty introduces Risk



With a bond, your final payout (Principal) is fixed. With a stock, there is no maturity date and **no guaranteed return of capital**.

- ▶ **The Liquidity Reality:** To get your money back, you must sell your shares to the open market at the current market price.
- ▶ **Price Risk (Volatility):** This market price fluctuates every single second.

Why does the price fluctuate?

Price changes are simply the market constantly updating its *expectations*.

Price Risk = Changing Expectations of Future Dividends

Pricing

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We established that owning assets like stocks exposes you to **Price Risk**.

What is Hedging?

- ▶ Hedging is essentially taking out **insurance** on your portfolio.
- ▶ It is a strategy designed to offset or minimize the risk of adverse price movements.
- ▶ *Goal:* We willingly give up some potential upside to strictly limit our potential downside.

The Mechanism

To hedge, you take an opposite position in a related asset. If your main asset loses value, the hedge gains value, mathematically canceling out the loss.

How do we practically eliminate the risk of a price changing tomorrow?

By agreeing on the price today.

- ▶ **The Concept:** A Futures contract is a legally binding agreement to buy or sell an asset at a predetermined price, at a specific time in the future.
- ▶ **Example (The Farmer):** A wheat farmer fears the price of wheat will crash by harvest time. He sells a Futures contract today at ₹2,000/ton.
- ▶ **The Result (Risk = 0):** No matter what happens to the market in the future, the farmer's selling price is mathematically locked. The price risk is gone.

If Futures allow us to lock in prices, **who decides what that locked-in price should be?**

This is where **Arbitrage** fits in. It acts as the "gravity" of financial markets.

- ▶ **Definition:** Arbitrage is the act of making a *risk-free profit* with zero net investment by exploiting price mismatches.
- ▶ **The Law of One Price:** If two identical assets have different prices, an arbitrageur will buy the cheap one and sell the expensive one until the prices equalize.

Arbitrage drives Pricing

We don't guess the price of a Futures contract. We use math to find the exact price where **no arbitrage is possible**. That is the "Fair Price."

What if you need to hedge a complex risk, but a specific Futures or Options contract doesn't exist in the market?

The Solution: Replication

- ▶ Mathematical finance proves that you can **replicate** the payoff of any derivative by dynamically trading a mix of the underlying Stock and a Risk-Free Bond.
- ▶ **The Connection to Arbitrage:** If your synthetic "Stock + Bond" portfolio produces the exact same cash flows as the complex derivative...
- ▶ **The Result:** By the Law of No-Arbitrage, the price of the complex derivative *must* equal the cost of setting up the replicating portfolio!

Pricing

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We have seen that **Hedging** (using derivatives like Futures) is a precise tool to neutralize specific price risks. But it often costs money or limits upside.

Is there another way to mitigate risk without buying derivatives?

Yes: Diversification.

- ▶ **The Core Idea:** "Don't put all your eggs in one basket."
- ▶ **The Mechanism:** Instead of holding one risky stock, we hold a *portfolio* of many different assets (tech stocks, bank stocks, bonds, gold).
- ▶ **The Result:** When one asset drops in price, another may rise. The individual price risks mathematically cancel each other out over time.

In 1952, Harry Markowitz transformed diversification from a simple "rule of thumb" into a rigorous mathematical science.

The Mathematical Breakthrough:

- ▶ Markowitz proved that the risk of a portfolio is **not** just the average risk of the individual stocks.
- ▶ The true risk depends heavily on **Correlation** (Covariance). How do the assets move *relative to each other*?

The Optimization Problem

We can use linear algebra to construct the **Efficient Frontier**:

1. For a given level of Risk $\rightarrow$ Maximize the Expected Return.
2. For a given Expected Return $\rightarrow$ Minimize the Risk (Variance).

*We no longer look at assets in isolation. We only care about how an asset changes the risk-return profile of the **entire portfolio**.*

Earlier, we priced a Bond assuming the cash flows were 100% guaranteed. But in the real world, companies and governments can go bankrupt.

What is Credit Risk?

- ▶ It is the probability that a counterparty will default on their debt obligations (fail to pay the coupon or principal).
- ▶ *The Rule:* If cash flows are less certain, the asset is worth less today.

Pricing Credit Risk: The Credit Spread

Investors punish risky bonds by demanding a higher discount rate (r).

Risky Yield = Risk-Free Rate + Credit Spread

The higher the chance of default, the wider the spread, and the lower the bond price.

We used Futures to hedge against *Price Risk*. But how do we hedge against *Default Risk*?

The Solution: Credit Default Swaps (CDS)

- ▶ A CDS is essentially an **insurance policy** on a bond.
- ▶ **How it works:** You pay a small regular premium to a CDS seller. If the bond issuer goes bankrupt, the CDS seller pays you the full principal you lost.
- ▶ **The Result:** By owning the Bond + buying a CDS, you have mathematically synthesized a risk-free asset! (Replication at work).

Credit Risk perfectly mirrors Market Risk. We use the exact same three mathematical tools to manage it:

- ▶ **1. Pricing (Valuation):** We use probability models to calculate the *Probability of Default (PD)* and *Loss Given Default (LGD)* to find the fair price of a corporate bond.
- ▶ **2. Hedging (Derivatives):** We trade Credit Default Swaps (CDS) to transfer the risk of bankruptcy to someone else, enforcing no-arbitrage bounds.
- ▶ **3. Portfolio Construction (Diversification):** Just like stocks, we don't lend to just one company. We build diversified portfolios of loans (like CLOs) so that a single corporate bankruptcy does not destroy the fund.

We can model markets (Market Risk) and defaults (Credit Risk), but what about the internal plumbing of the financial system?

What is Operational Risk?

- ▶ The risk of loss resulting from inadequate or failed internal processes, people, systems, or external events.
- ▶ *Examples:* A trader entering a wrong extra zero (fat-finger error), a cyberattack, a server crash, or massive fraud (e.g., Nick Leeson at Barings Bank).

The Mathematical Challenge

- ▶ Market and Credit risk have abundant, structured historical data.
- ▶ Operational Risk is characterized by **Low Frequency, High Severity** events (black swans).
- ▶ We must use extreme value theory and advanced statistical distributions to model things that rarely happen but cause catastrophic damage.

The Role of Quants

Who builds and applies these mathematical frameworks in the real world?

Quantitative Analysts (Quants)

Specialists operating at the intersection of:
Mathematics $\times$ **Computer Science** $\times$ **Finance**

Their core responsibilities include:

- ▶ Pricing exotic derivatives and structuring hedges.
- ▶ Designing high-frequency algorithmic trading strategies.
- ▶ Optimizing massive institutional portfolios.
- ▶ Building predictive models for credit defaults and fraud.

The Quant Ecosystem: Specializations

The "Quant" title is broad. In modern finance, the role is highly specialized:

Quant Researcher (*The Architects*)

Invent and rigorously test new mathematical models for pricing, alpha generation, and risk forecasting.

Quant Developer (*The Engineers*)

Translate pure mathematical models into ultra-fast, scalable, and robust production code (C++, Python).

Quant Trader / Strat (*The Executors*)

Deploy these quantitative models directly in the market to execute trades and capture profit.

Risk Quant (*The Guardians*)

Quantify, monitor, and stress-test the bank's exposure to Market, Credit, and Operational risks.

The Role of AI in Finance

Classical mathematical finance successfully solved **Pricing, Hedging, and Portfolio Optimization**. However, these models rely on rigid assumptions:

Classical Assumptions:

- ▶ Parameters (like volatility) are stable.
- ▶ Historical data strictly informs the future.
- ▶ Investors behave rationally.

Market Reality:

- ▶ Market regimes change rapidly.
- ▶ Correlations fluctuate wildly during crises.
- ▶ Participant behavior constantly evolves.

The Challenge

Static mathematical models struggle to survive in dynamic, evolving environments.

Many financial challenges are fundamentally **prediction and pattern discovery** problems that are too complex for explicit classical formulas.

Representative ML Applications:

- ▶ **Credit:** Predicting defaults and scoring borrowers.
- ▶ **Risk:** Forecasting volatility and detecting fraud.
- ▶ **Alpha:** Discovering hidden trading signals.

The ML Workflow

Raw Data → Feature Engineering → Learning Model → Prediction

AI emerges not because mathematical finance failed, but because modern finance is flooded with new, unstructured data types.

Structured (Numerical)

- ▶ Asset Prices & Volumes
- ▶ Order Books
- ▶ Economic Indicators

Unstructured (Text & NLP)

- ▶ Earnings Transcripts & News
- ▶ Regulatory Filings
- ▶ Social Media Sentiment

The Key Idea:

Markets react just as strongly to the information contained in **language** as they do to numerical data.

The future of the industry is a close, interactive synthesis of both disciplines.

Traditional Finance Provides:

- ▶ Valuation Principles
- ▶ Hedging Frameworks
- ▶ Risk Management Limits

AI Provides:

- ▶ Pattern Discovery
- ▶ High-speed Forecasting
- ▶ Language Understanding

Final Takeaway:

The most successful practitioners will use **AI** to find the signals, and **Mathematical Finance** to safely execute the decisions.

Course Roadmap

Course Roadmap: The Journey Ahead

Curriculum bridges the gap between foundational economic theory and artificial intelligence.

Phase I: Mathematical Finance

1. Time Value of Money & Instruments
2. Equities, Bonds, Futures, & Options
3. Principles of Pricing & Hedging
4. Portfolio Construction & Risk
5. Credit Risk & Structured Products

Phase II: The AI Frontier

6. Machine Learning in Finance
7. Algorithmic Trading
8. Financial NLP (Text as Data)
9. Generative AI & Agentic Systems

*We move systematically from classical guarantees
to modern, AI-enabled uncertainty.*

1. Mathematical Finance

- ▶ Probability & Statistics
- ▶ Numerical Methods
- ▶ Convex Optimization
- ▶ Stochastic Processes
- ▶ Differential Equations

2. Artificial Intelligence

- ▶ **Machine Learning (ML):** Regression, & Clustering
- ▶ **Deep Learning (DL):** Neural Networks & Sequences
- ▶ **Reinforcement Learning (RL):** Sequential decision-making

The Takeaway

Combining **mathematical rigor** with **data-driven learning**.

Comfortably navigate beyond traditional FDs and make objective investment decisions

Core Capabilities:

- ▶ **Market Intuition:** Understand how modern financial markets are organized and instruments behave.
- ▶ **Quantitative Rigor:** Reason mathematically about risk and return, understand of pricing and hedging.
- ▶ **AI Discernment:** Know exactly *where* machine learning adds value, and *where* classical financial reasoning remains indispensable.

It is a course about mastering the intersection of three disciplines:

Financial Reasoning

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Mathematical Modelling

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Artificial Intelligence

Welcome to AI in Finance.

**Prof. Ganesh
Ghalme**

**Prof. Raju
Chinthalapati**

**Dr. Easwar
Subramanian**